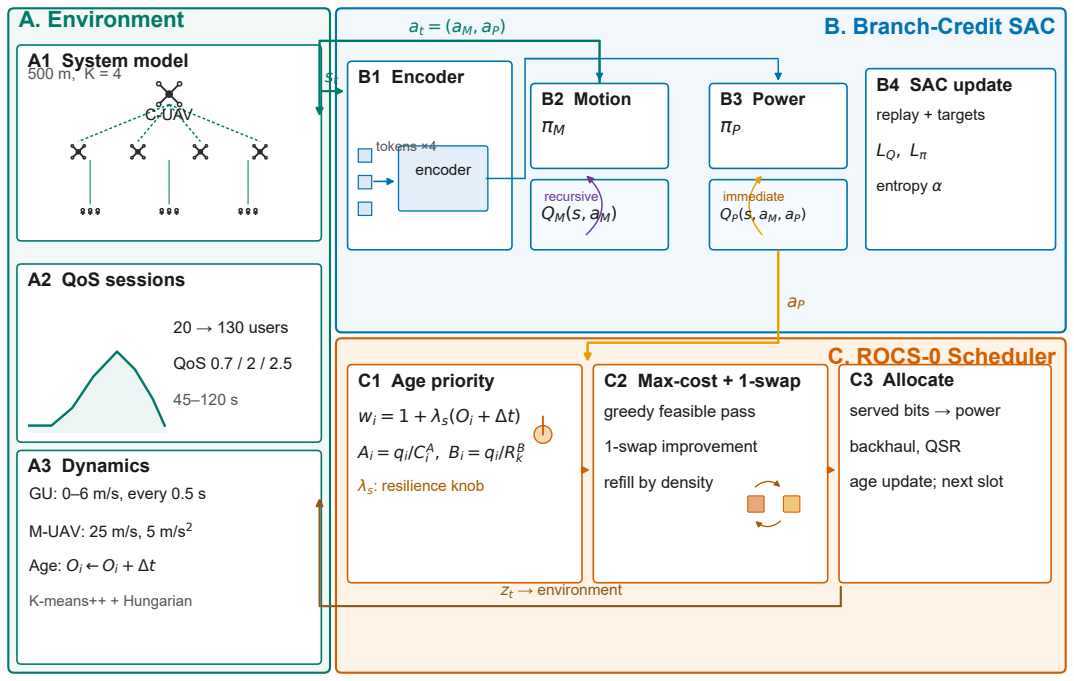

A. Environment
A1 System model
500 m, K = 4
C-UAV
A2 QoS sessions
20 → 130 users
QoS 0.7 / 2 / 2.5
45–120 s
A3 Dynamics
GU: 0–6 m/s, every 0.5 s
M-UAV: 25 m/s, 5 m/s²
Age: $O_i \leftarrow O_i + \Delta t$
K-means++ + Hungarian
$a_t = (a_M, a_P)$
s_t
B. Branch-Credit SAC
B1 Encoder
tokens ×4
encoder
B2 Motion
π_M
recursive
$Q_M(s, a_M)$
B3 Power
π_P
immediate
$Q_P(s, a_M, a_P)$
B4 SAC update
replay + targets
L_Q, L_π
entropy α
a_P
C. ROCS-0 Scheduler
C1 Age priority
$w_i = 1 + \lambda_s(O_i + \Delta t)$
$A_i = q_i/C_i^A$, $B_i = q_i/R_k^B$
λ_s: resilience knob
C2 Max-cost + 1-swap
greedy feasible pass
1-swap improvement
refill by density
C3 Allocate
served bits → power
backhaul, QSR
age update; next slot
z_t → environment